

DUPAM HARSHITH

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CAREER OBJECTIVE

An academically exceptional and highly driven Chemical Engineering student with strong foundations in software development, computational logic, and algorithmic problem-solving. Effectively combines data structures, system automation, and relational database paradigms to engineer high-performance terminal applications and interactive software systems. Seeking to leverage strong programming capabilities in Python and C to tackle complex engineering, process modeling, and data-driven challenges.

EDUCATION

Chaitanya Bharathi Institute of Technology (CBIT) — *Hyderabad, India*

Bachelor of Technology (B.Tech) in Chemical Engineering

Expected Graduation: 2029 | Current Standing: First Year

- **Academic Performance:** CGPA / SGPA: 10.0 / 10
- **Core Computational Context:** Strong focus on computational engineering matrices, relational workflows, and algorithm design structures.

TECHNICAL SKILLS

- **Programming Languages:** Python, C , Html, CSS
- **Core Competencies:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Memory Safety, Real-Time Loop Architectures
- **Database & Storage:** SQLite, Relational Database Design, File I/O Streams
- **Tools & Game Engines:** Git, GitHub, Godot Engine, Visual Studio Code, Pygame

TECHNICAL PROJECTS

SQL-Driven Interactive Quiz Engine |

Python & SQLite

- Designed and deployed a localized quiz application mapping multi-subject STEM questions spanning complex mathematical, chemical, and physical laws.
- Implemented a relational database schema using sqlite3 featuring robust data normalization, bulk data injection (executemany), and automated safe schema rebuild steps.

- Engineered randomized, dynamic question fetching by offloading processing overhead straight to the database engine utilizing specialized SQL clauses (ORDER BY RANDOM() LIMIT).

Space Defenders Arcade Game |

Python & Pygame

- Programmed a 2D vertical scrolling arcade space shooter running at a stable 60 FPS frame rate.
- Engineered custom frame time delta throttling and high-resolution hardware clock polling (time.perf_counter) to handle exact bullet weapon reload cooldown delays.
- Integrated memory-safe execution sweeps utilizing active list comprehensions to clean out off-screen entities and avoid memory accumulation debt.
- Managed cross-session game-state saving using file handling output streams to store persistent high score data locally.

Algorithm Toolkit Library |

Python & Low-Level C

- Authored a complete programmatic reference collection containing 5 core searching and 5 classical sorting algorithms entirely from scratch.
- Successfully translated abstract mathematical concepts (including recursive divide-and-conquer structures for Merge/Quick Sort and exponential boundary steps for Jump Search) into optimized, error-free source logic.
- Analyzed and documented codebases with descriptive docstrings detailing targeted worst-case and best-case Time Complexities ($O(n \log n)$, $O(n)$, etc.).

Mini RPG Game (In Progress) |

Godot Engine

- Designing a 2D role-playing game utilizing the Godot platform to manage interactive sprite vectors and structural object layouts.
- Architecting modular script relationships to direct dynamic player movement arrays and state-driven experience progression mechanics.

CERTIFICATIONS & EXTRACURRICULARS

- **NPTEL Professional Certification – Cloud Computing:** Deeply studied architectural deployment models, SLO/SLA metrics, CAP Theorem properties, hypervisor computing logic, HDFS, MapReduce, Docker, and OpenStack systems.
- **Competitive Programming:** Active participant in student Hackathons, coding marathons, and structural algorithmic challenges.